

# Lesson 3: Is a force real?

The Brake · Newton's first and second laws · 6 periods of 50 minutes

Grade 11 scientific track (adaptable to Grade 10 and to equivalent grades in other Arab countries). Physics in English or French; discussion may be in Arabic. The student lesson is free at [sababphysics.com](https://sababphysics.com): Lessons, Grade 11, Lesson 3.

## Learning objectives

- Tell apart describing a motion (kinematics) and explaining it with its causes (dynamics). (A1)
- Determine a mass as the constant ratio force ÷ acceleration from experimental data. (A2)
- State Newton's second law in vector form,  $\Sigma \mathbf{F} = m \cdot \mathbf{a}$  (vectors in bold), with  $\mathbf{a}$  in the direction of the net force. (A3)
- Represent the forces on a body and project the law on the direction of motion to calculate an acceleration. (A4)
- Explain the braking passenger using only real forces, exerted by an identified object. (C1)
- Argue whether a force is real or a description, answering the strongest opposite view. (C2, C3)

## Competencies of the 2026 framework

| Competency                      | Where it is developed                             |
|---------------------------------|---------------------------------------------------|
| Critical thinking               | Steps 2 and 6: position, argument, position shift |
| Inquiry                         | Steps 3 and 4: collect data, find the law         |
| Digital                         | Virtual lab and data table                        |
| Communication and collaboration | Community of inquiry, written reflection          |
| Creativity and innovation       | Step 5: Arduino lab or egg-passenger challenge    |

## The six steps at a glance

- |                      |                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------|
| <b>1 Hook</b>        | Film "The Brake" and the taxi simulation. Students predict what pushes the passenger forward.      |
| <b>2 Question</b>    | "Is a force real?" Four thinkers, four answers. Students take a position.                          |
| <b>3 Investigate</b> | Virtual cart: vary the hanging mass and the cart mass, collect accelerations, compute $F \div a$ . |
| <b>4 Model</b>       | Test four models against the data; Newton's second law emerges. Back to the taxi.                  |
| <b>5 Make</b>        | Arduino lab (measure a real acceleration) or the egg-passenger challenge.                          |
| <b>6 Reflect</b>     | Position shift, concept ladder, written reflection.                                                |

## Before you start: prerequisites

From Chapter 7 and earlier grades. Check them in period 1; if many students are unsure, add a 15-minute review.

- Vectors: add two vectors, and project a vector on two perpendicular axes.
- Velocity and acceleration vectors;  $a = \Delta v \div \Delta t$ .
- Uniformly varied rectilinear motion:  $v = a \cdot t + v_0$  and  $x = \frac{1}{2} \cdot a \cdot t^2 + v_0 \cdot t + x_0$ .
- Forces from earlier grades: weight  $P = m \cdot g$ , normal reaction, tension, friction, and how to draw them.

## Timeline: six 50-minute periods

|          |                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| P1 · 0'  | Hook: play the film and the taxi animation on the projector. Students predict individually, then vote.                                          |
| P1 · 15' | Describe or explain? Students sort four sentences; name kinematics and dynamics only afterwards.                                                |
| P1 · 35' | Prerequisite check: project a force on two axes; read a velocity and an acceleration from a table.                                              |
| P2 · 0'  | Community of inquiry (circle): "Is a force real?" Read the four thinkers. Roles: facilitator, questioner, summariser, skeptic.                  |
| P2 · 35' | Students write their position and one reason, then answer the strongest argument from the other side (step 2). Homework: 3 lab runs on a phone. |
| P3 · 0'  | Investigate (step 3): groups run the virtual lab, vary the force and the mass, and fill the table, including the $F \div a$ column.             |
| P3 · 35' | Groups compare their data: for one cart, $F \div a$ stays constant and equals its mass.                                                         |
| P4 · 0'  | Model (step 4): test the four candidate models against the data, then name the law $\Sigma F = m \cdot a$ .                                     |
| P4 · 25' | Explain the taxi with real forces only. The crate: represent the four forces and project on the direction of motion ( $F - f = m \cdot a$ ).    |
| P5 · 0'  | Make (step 5): half the class runs the Arduino lab, half builds the egg-passenger cart.                                                         |
| P5 · 40' | Groups swap findings: when the measurement and the theory disagree, why?                                                                        |
| P6 · 0'  | Exam-style practice: the quiz of the lesson, corrected together.                                                                                |
| P6 · 30' | Reflection and position shift (step 6). Students send their reflection to the class group.                                                      |

## The four thinkers (step 2)

### Aristotle

4th century BCE

A moving body needs a mover. Without a continuing push, motion stops. For a thrown stone, the air keeps pushing it (*Physics*, VII and VIII).

### Ibn Sina (Avicenna)

11th century

A thrown body receives an inclination (mayl) from the thrower that keeps it moving until resistance wears it away. In a void, it would never stop (*The Book of Healing*).

### Isaac Newton

1687

A body keeps its state of rest or of straight, uniform motion unless a net force acts. Force changes motion: net force =  $m \cdot a$ .

### Ernst Mach

19th century

Mass and force are defined by the accelerations bodies give each other. Force is a name for mass  $\times$  acceleration, not a hidden cause (*The Science of Mechanics*, 1883).

*Short summaries for discussion, not quotations.*

## Common misconceptions and how to challenge them

**Motion needs a force.**

Ask: what keeps a puck sliding on ice? Show the virtual cart with no friction.

**A moving object "has" a force that runs out.**

Compare with Ibn Sina's mayl, then ask what the data from step 3 shows.

**Heavier objects always accelerate less, whatever the force.**

Point to runs where the force changes with the mass.

**The forward "push" felt when braking is a real force.**

Ask: what object exerts it? No object, no force: the passenger keeps moving while the taxi slows.

## Make step: Arduino lab

- Per group: Arduino Uno, HC-SR04 distance sensor, 4 jumper wires, toy car, string, small water bottle.
- Students paste the Serial Monitor data; the platform fits  $x(t)$  and compares  $a$  with the theory.
- No kits? Use the Wokwi online simulator or the built-in sample data.
- Discussion: when theory and measurement disagree, which one is "wrong"?

## Make step: egg-passenger challenge

- Recycled materials only: bottles, cardboard, sponges, tape.
- Teams predict with the on-screen calculator, build, test, then improve once.
- Score: egg intact (3), prediction matched the test (2), design journal uses  $F = m \cdot a$  (3), poster (2).
- Citizenship link: display the posters in real service taxis.
- Safety: raw eggs, low ramps, clean up spills at once.

## Facilitation moves

- "What do you mean by...?" (clarify)
- "How do you know?" (evidence)
- "Can anyone give a counter-example?"
- "What would Ibn Sina say to Aristotle?"
- Aristotle said the air keeps pushing a thrown stone. Ask: what would happen in a vacuum?
- Do not give the answer before step 4.

## Optional: a second simulation

For step 3, teachers can also use **PhET · Forces and Motion: Basics**

(phet.colorado.edu/en/simulations/forces-and-motion-basics). Free, in Arabic, French and English; students can turn friction on and off and show the sum of forces. Credit: PhET Interactive Simulations, University of Colorado Boulder, licence CC BY-NC 4.0.

## Assessment rubric: written argument (steps 2 and 6)

| Criterion        | 1        | 2                       | 3                            |
|------------------|----------|-------------------------|------------------------------|
| Claim            | Unclear  | Clear                   | Clear and nuanced            |
| Evidence         | None     | Daily-life example      | Uses own lab data            |
| Reasoning        | Restates | Links evidence to claim | Uses Newton's laws correctly |
| Counter-argument | Absent   | Mentioned               | Answered                     |

### Student worksheet · Lesson 3

Photocopy for each student

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

#### 1. Predict: what pushes the passenger forward when the taxi brakes?

- ☐ A The braking pushes the passenger forward.
- ☐ B Nothing pushes them. They keep moving while the car slows down.
- ☐ C The car's motion stays in the passenger and carries him forward.
- ☐ D The air inside the car pushes them.

#### 2. My data (step 3)

| Run | Hanging mass (g) | Cart mass (g) | Net force F (N) | Acceleration a (m/s <sup>2</sup> ) | F ÷ a (kg) |
|-----|------------------|---------------|-----------------|------------------------------------|------------|
| 1   |                  |               |                 |                                    |            |
| 2   |                  |               |                 |                                    |            |
| 3   |                  |               |                 |                                    |            |
| 4   |                  |               |                 |                                    |            |
| 5   |                  |               |                 |                                    |            |

#### 3. My position: is a force a real thing, or only a description?

A real thing

Only a description



Mark B (before, step 2) and A (after, step 6) on the line.

#### 4. Exit ticket

A passenger lurches forward when a service taxi brakes suddenly. Explain why, using only real forces (forces exerted by an object).

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## Assessment rubric: the Make challenge (step 5)

Four criteria, four levels. Total out of 16. Level 3 is the expected standard.

| Criterion           | 1 • Beginning                               | 2 • Developing                                    | 3 • Secure                                                 | 4 • Excellent                                                                         |
|---------------------|---------------------------------------------|---------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Physics used        | No physics in the design or journal         | Names inertia or $F = m \cdot a$ without using it | Uses $F = m \cdot a$ to choose the stopping distance       | Calculates the force on the egg and explains why a longer stopping distance lowers it |
| Design              | Cart does not roll or does not hold the egg | Holds the egg, no crumple zone                    | Crumple zone and belt from recycled materials              | Design improved once after a test, with the reason written down                       |
| Data and prediction | No prediction or measurement                | Prediction made, not compared with the test       | Prediction compared with the test result                   | Ramp height and stopping distance measured; the gap with the prediction is explained  |
| Communication       | No poster or journal                        | Poster without physics                            | Poster with one force diagram and one number from the test | Clear bilingual message that a taxi passenger would understand                        |

## Answer keys

**Step 2 • Describe or explain?** (1) describes (kinematics). (2) explains with a cause (dynamics). (3) describes (kinematics). (4) explains with a cause (dynamics).

**Step 4 • The crate:**  $F - f = m \cdot a$ . P and N are perpendicular to the motion, so their projections are zero ( $N = P$ ); F points along the motion (+), f against it (-).

**Step 4 • Exam-style quiz**

1. **B: 6,000 N** (A3)  $a = \Delta v \div \Delta t = 20 \div 4 = 5 \text{ m/s}^2$ , so  $F = m \cdot a = 1,200 \times 5 = 6,000 \text{ N}$ .

2. **C: 3 m/s<sup>2</sup>** (A4) Net force =  $10 - 4 = 6 \text{ N}$ , so  $a = 6 \div 2 = 3 \text{ m/s}^2$ .

3. **C: 12 m/s<sup>2</sup>** (A4) Forces: P (down), N (up), F (forward), f (backward). P and N cancel vertically. Along the motion:  $F - f = m \cdot a$ , so  $a = (200 - 80) \div 10 = 12 \text{ m/s}^2$ .

4. **B: It keeps moving at constant velocity because the net force is almost zero.** (C1) First law: with no net force, velocity stays constant. The stick's force acted only during contact.

5. **B: It keeps increasing, by the same amount every second.** (A3) A constant net force gives a constant acceleration ( $\Sigma F = m \cdot a$ ), so the speed grows by the same amount every second:  $v = a \cdot t + v_0$ . A force changes the speed; it does not set it. On a smooth track, the speed never levels off.

6. **C: His weight P and the push N of the seat, equal and opposite.** (A4) At rest,  $a = 0$ , so  $\Sigma F = 0$ : the forces cancel, they do not disappear. The Earth pulls him down ( $P = m \cdot g \approx 686 \text{ N}$ ) and the seat pushes him up with  $N = 686 \text{ N}$ . A seat can push: it is squeezed a little, like a spring.

7. **C: Neither: both accelerate at 2 m/s<sup>2</sup>.** (A2)  $a = F \div m$ : for A,  $2 \div 1 = 2 \text{ m/s}^2$ ; for B,  $4 \div 2 = 2 \text{ m/s}^2$ . What matters is the ratio of force to mass, not the mass alone or the force alone.

**Step 6 • Exit ticket (C1), model answer:** When the driver brakes, the road pushes backward on the tyres (friction) and the taxi slows down. No object pushes the passenger forward: by inertia he keeps moving at the same speed, so he moves forward relative to the slowing taxi. To slow him down, an object must push or pull him backward. With a seat belt, the belt pulls him backward and he slows down with the car. Without a belt, as in the film, the friction of the seat on his body is too weak to slow him as fast as the car, so he keeps moving forward until something in front stops him: here, the back of the front seat pushes backward on his hand. *Full marks: names the road as the object that slows the taxi and an object that slows the passenger (the belt, or the front seat if he wears none), uses inertia, and states that there is no forward force.*

## Discussion guide: the community of inquiry (step 2)

### Running the circle

- Chairs in a circle. Read the big question aloud, then give two minutes of silent writing.
- Give the roles: facilitator, questioner, summariser, skeptic. Change them each session.
- Rules: one voice at a time; answer the previous speaker; disagree with ideas, not people.
- The teacher asks questions and does not give answers. 20 to 25 minutes is enough.

### Closing without a moral

- The summariser states the positions heard. Name what the class agrees on and what stays open.
- Each student writes their position for now, with one reason.
- Do not announce a right philosophical answer or a moral. The physics answer comes from the data in step 4.

### Sentence starters

| English                               | Français                                       | Arabic                      |
|---------------------------------------|------------------------------------------------|-----------------------------|
| I agree with ... because ...          | Je suis d'accord avec ... parce que ...        | أُتفق مع هذا الرأي لأن      |
| I disagree with ... because ...       | Je ne suis pas d'accord avec ... parce que ... | لا أوافق مع هذا الرأي لأن   |
| Can you give an example?              | Peux-tu donner un exemple ?                    | هل يمكنك أن تعطي مثالاً؟    |
| What would make you change your mind? | Qu'est-ce qui te ferait changer d'avis ?       | ما الذي قد يجعلك تغير رأيك؟ |

## Differentiation

### Support

- Give the data table half filled in; vary only the force first, then only the mass.
- In the circle, pair a confident speaker with a quiet one, and give the sentence starters.
- Let students answer the exit ticket aloud first, then write it.
- Start the quiz with questions 1 to 3, with a calculator.

### Extension

- Plot F against a for one cart: the slope is its mass.
- With friction on, use  $F - f = m \cdot a$  to work out the friction force from the data.
- Compare Ibn Sina's mayl with the modern idea of momentum ( $m \cdot v$ ).
- Preview Lesson 5: what changes for the crate if the floor is a slope?

## Materials and safety

- Step 3, simple version: toy car, string, small bottle of water, tape measure, phone to film. Put a cushion under the bottle.
- Arduino lab: Arduino Uno, HC-SR04 sensor, 4 jumper wires, breadboard, toy car, string, 50–100 g hanging mass. Put a stopper at the end of the table.
- Egg challenge: raw eggs, recycled bottles, cardboard, sponges, tape, a low ramp. Wash hands after handling raw eggs; clean spills at once.
- Phone acceleration apps in a real car: only as a passenger, phone held in place, never while driving.